

# SMART BIO AIR - REPORTING SYSTEMS

AI-driven Bioreactor & Indoor Air Quality Platform

## DAILY DIAGNOSTIC SUMMARY

Generated on: 2026-07-31 21:08:43

| 1. CORE TELEMETRY                |                                  |
|----------------------------------|----------------------------------|
| Algae Temperature: 21.8 C        | Photosynthesis Light: 1118.2 Lux |
| Green Index (density): 0.86      | Estimated Biomass: 0.175 g/L     |
| Atmospheric Pressure: 1013.2 hPa | Altitude: 150.0 m                |
| Air Quality Score (MQ135): 74.0% | Motor Status: ON (60.0% speed)   |

| 2. ANALYTICS & DIAGNOSTICS     |                                  |
|--------------------------------|----------------------------------|
| Algae Health Index: 62.0%      | Photosynthesis Efficiency: 42.6% |
| Stress Index: 8.9%             | Comfort Index: 99.2%             |
| Environmental Stability: 99.7% | Growth Velocity: 0.00000 G/L/hr  |
| Motor Life (RUL): 4879.4 hrs   | Next Bio-Cleaning Due: 10.0 days |

| 3. ACTIVE ALERTS & ANOMALIES                                                        |
|-------------------------------------------------------------------------------------|
| [WARNING] Algae stress warning. Health index decreased to 62.0%. Inspect nutrients. |

| 4. AI RECOMMENDATIONS & ACTIONS                                                                                                                                                                                                                                                                                                         |
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| 1. ["Increase light intensity if green index is low and light is below 500 lux.", "Inspect pump motor speed if water flow rate deviates from target value.", "Verify algae culture density; consider partial harvesting to prevent light blocking.", "Check temperature controls; ensure water stays between 20\u00b0C and 28\u00b0C."] |

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| 5. SCIENTIFIC RESEARCH NOTES                                                                                                                                                                                                                                                                       |
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| Research Note: The bioreactor indicates stable growth conditions. Photosynthesis is active as indicated by positive correlations between Light Lux and estimated biomass. Gas concentrations are currently within normal parameters. The green index trend shows incremental density accumulation. |
| (AI System Notice: Gemini API was offline or unauthorized. Displaying automated diagnostics.)                                                                                                                                                                                                      |